

Delay distributions limit the identifiability of rapid variations in epidemics

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Abstract

Public-health surveillance systems rely on downstream indicators to infer latent infection incidence, but delays and observation noise provide only an indirect and temporally distorted view of the underlying epidemic process. Reconstructing upstream epidemic trajectories from these observations is therefore an ill-posed inverse problem, in which different reconstruction assumptions may produce different trajectories that remain consistent with the observed data. Here, we develop a general spectral framework that quantifies the statistical distinguishability of candidate upstream trajectories under delayed and noisy observations. We show that epidemiological delay distributions impose a frequency-dependent temporal resolution limit on epidemic surveillance, fundamentally constraining the distinguishability of rapid upstream variation. This limitation propagates to epidemiological inference, making some quantities substantially more sensitive to reconstruction assumptions than others and rendering distinct event-impact profiles difficult to distinguish from downstream observations.

Introduction

Public-health surveillance systems rely on downstream indicators such as confirmed cases, hospitalizations, and deaths to infer the trajectory of infection incidence[20, 21, 28]. However, these observables do not measure infection incidence directly. Instead, they arise only after infected individuals progress through a sequence of biological, behavioral, and reporting steps—symptom onset, testing or care-seeking, laboratory processing, and administrative reporting—each introducing delays, partial ascertainment, and additional sources of noise[12, 9, 18].

Consequently, surveillance data provide only an indirect and temporally distorted view of the underlying epidemic process. Because rapid changes are progressively attenuated before they appear in downstream observations, surveillance systems have a finite temporal resolution: sufficiently short-lived variations in transmission may no longer remain statistically distinguishable, limiting their identifiability regardless of the inference algorithm subsequently applied. This information-loss process is illustrated schematically in Fig. 1, where distinct upstream trajectories become nearly indistinguishable after delay filtering and observation noise.

A wide range of statistical and mechanistic approaches have been developed to reconstruct latent epidemic trajectories from delayed observations, including back-projection[4], deconvolution[15], Bayesian latent-incidence reconstruction[6], and renewal-based inference[8, 16]. Despite these methodological advances, reconstructing latent incidence from delayed observations remains an ill-posed inverse problem because multiple upstream trajectories may be consistent with the same downstream observations[28]. Stable inference therefore requires regularization, prior distributions, or mechanistic constraints that select among multiple trajectories compatible with the observed data [28, 4, 24]. These assumptions appropriately quantify uncertainty within a specified model, but conditional uncertainty is not equivalent to observation-level identifiability: a narrow posterior does not imply that the observations uniquely determine the inferred trajectory. When the observations weakly constrain fine-scale temporal variation, different reconstruction assumptions can yield substantially different upstream dynamics while remaining equally consistent with the observed downstream data[28, 6].

Related questions have been investigated within several complementary theoretical traditions. Within epidemiology, the primary question is how latent incidence should be reconstructed from delayed observations under specified epidemiological assumptions[6, 28]. In control theory and state-space modeling, observability

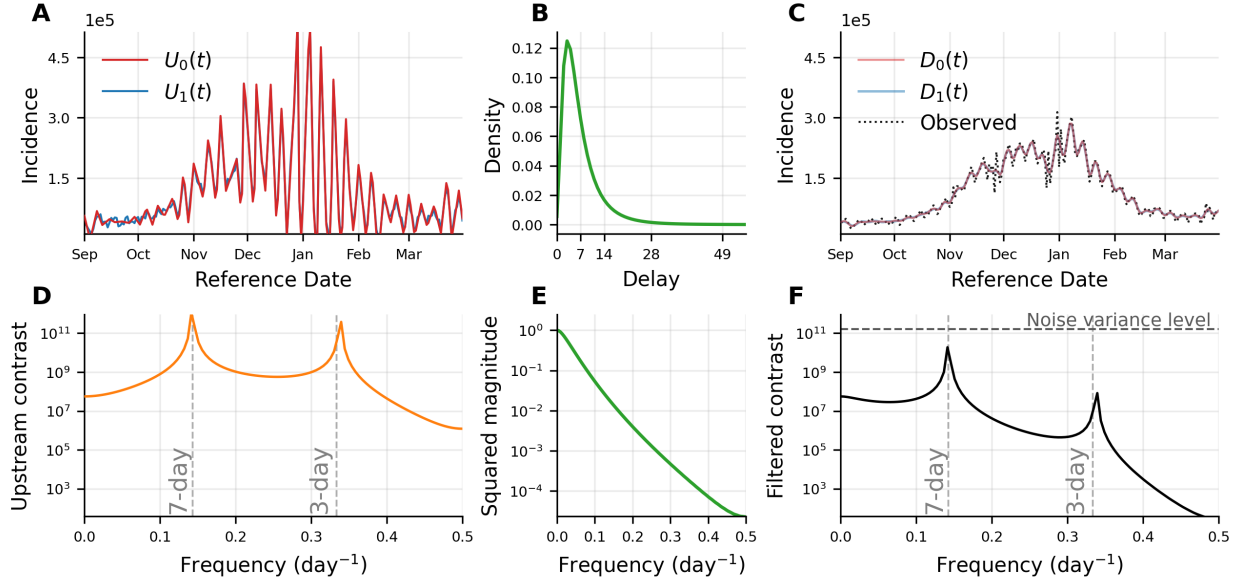


Figure 1: Delayed observation attenuates temporal differences between upstream epidemic trajectories. (A) Two illustrative candidate upstream incidence trajectories, constructed through the data-informed procedure described in Appendix S1. (B) Published COVID-19 symptom-onset-to-report delay distribution used to map upstream incidence to downstream observations [32]. (C) Corresponding delay-convolved trajectories, $D_0(t)$ and $D_1(t)$, together with observed U.S. daily confirmed-case incidence from the JHU-CSSE data stream [10]. (D) Frequency-specific contrast between the upstream trajectories, with frequencies corresponding to 7- and 3-day periods indicated. (E) Squared magnitude of the delay-kernel frequency response. (F) Downstream spectral contrast after delay filtering, shown relative to the effective observation-noise level. Delay filtering preferentially suppresses high-frequency differences, potentially rendering distinct upstream trajectories statistically indistinguishable. Construction details are provided in Appendix S1.

characterizes whether hidden states are recoverable from noisy measurements under a specified dynamical system, while filtering provides algorithms for estimating those states.[23, 26]. In much of the classical observability literature, the central question is whether latent states can be reconstructed under specified dynamical and observation models. Statistical detection theory instead asks whether competing signals or hypotheses can be distinguished in the presence of noise and formulates likelihood-based measures of detectability that underpin ideal-observer analyses in communication systems, signal processing, and medical imaging[17, 2, 1]. Although these traditions respectively emphasize epidemic reconstruction, latent-state recoverability, and statistical distinguishability, none directly addresses the central question: given realistic delay distributions and observation noise, which temporal differences between candidate upstream epidemic trajectories remain statistically distinguishable in the downstream observations? Answering this question requires linking the epidemic observation process to a likelihood-based measure of statistical distinguishability, while explicitly incorporating the effects of epidemic

delay and observation noise.

As illustrated in Fig. 1, the delay distribution acts as a low-pass filter whose frequency response strongly suppresses rapid temporal variation, transforming the upstream signal before it reaches the data stream used for inference[6, 27]. In the frequency domain, convolution has a natural interpretation in which delay kernels act as strong low-pass filters that preserve slow, long-term trends while sharply attenuating rapid fluctuations [29, 5]. Consequently, abrupt changes or high-frequency oscillations in the underlying epidemic process are smoothed out before appearing in surveillance data and may become statistically undetectable once observation noise is taken into account.

Here we develop a general spectral framework that quantifies the statistical distinguishability of candidate upstream epidemic trajectories under delayed and noisy observations. By combining the frequency response of epidemiologically realistic delay distributions with observation-model-specific noise, we derive a separability functional that determines whether differences between upstream trajectories remain recoverable in downstream

surveillance data. The framework yields explicit identifiability limits across common observation models and characterizes which temporal features are fundamentally recoverable, independent of the particular reconstruction method employed.

Methods

Spectral Representation for the Delay Model

The observation process considered throughout this study is summarized schematically in Fig. 1A–C. Let $U(t)$ denote the upstream process (e.g., latent infections) and $D(t)$ the downstream observable (e.g., confirmed cases or deaths). Let $g(\tau)$ be a nonnegative delay kernel supported on delays $\tau \geq 0$ and normalized so that $\int_0^\infty g(\tau) d\tau = 1$. A conversion rate $p \in (0, 1]$ represents the probability that an upstream event generates a downstream observation. A common example is the ascertainment rate, which describes the proportion of latent infections ultimately recorded as confirmed cases. Because testing- or reporting-related changes typically occur on coarse temporal scales [7, 11, 31] relative to the support of g , we treat p as locally constant.

The downstream process is modeled as

$$D(t) = (U * h)(t) + \varepsilon(t), \quad h(\tau) = p g(\tau), \quad (1)$$

where $*$ denotes convolution,

$$(U * h)(t) = \int_0^\infty U(t - \tau) h(\tau) d\tau,$$

and $\varepsilon(t)$ is a zero-mean residual satisfying $\mathbb{E}[\varepsilon(t) | U] = 0$. Hence,

$$\mathbb{E}[D(t) | U] = (U * h)(t).$$

For additive Gaussian observation models, $\varepsilon(t)$ corresponds directly to Gaussian noise. For count-based observation models such as Poisson or negative binomial sampling, this representation should instead be interpreted as a mean-plus-residual decomposition, where $\varepsilon(t)$ denotes deviations from the conditional mean.

Because convolution in the time domain corresponds to multiplication in the frequency domain, it is natural to examine (1) spectrally. Taking Fourier transforms yields

$$\widehat{D}(f) = \widehat{U}(f) \widehat{H}(f) + \widehat{\varepsilon}(f) = p \widehat{U}(f) \widehat{G}(f) + \widehat{\varepsilon}(f), \quad (2)$$

where $\widehat{X}(f)$ denotes the Fourier transform of $X(t)$. In particular, $\widehat{H}(f) = p \widehat{G}(f)$. When $p = 1$, equation (2) reduces to the delay-only setting without conversion loss. This frequency-domain representation highlights the attenuating effect of delay processes: for many epidemiological delay distributions, the magnitude $|\widehat{G}(f)|$ decreases with frequency, thereby attenuating

high-frequency upstream variation before it reaches the downstream process $D(t)$.

This framework extends naturally to multilayer delay structures in which several delay mechanisms act sequentially, such as incubation, testing, reporting, and hospitalization delays. Let $g_1, g_2, \dots, g_L$ denote normalized delay kernels associated with successive stages, and let $p_1, p_2, \dots, p_L$ denote the corresponding conversion rates. The overall conversion rate is $p_{\text{tot}} = \prod_{\ell=1}^L p_\ell$, while the overall delay kernel is given by the convolution $g_{\text{tot}} = g_1 * g_2 * \dots * g_L$. The effective impulse response therefore becomes $h_{\text{tot}}(\tau) = p_{\text{tot}} g_{\text{tot}}(\tau)$, and the downstream conditional mean satisfies

$$\mathbb{E}[D(t) | U] = (U * h_{\text{tot}})(t).$$

In the frequency domain, the corresponding transfer function factorizes as

$$\widehat{H}_{\text{tot}}(f) = \prod_{\ell=1}^L p_\ell \prod_{\ell=1}^L \widehat{G}_\ell(f). \quad (3)$$

Thus, sequential delay mechanisms compound their spectral attenuation multiplicatively, progressively suppressing high-frequency upstream variation before it reaches the downstream process.

Frequency-Domain Criterion for Upstream Identifiability

We consider identifiability of the upstream process by asking whether two candidate trajectories U_0 and U_1 can be statistically distinguished based on the downstream observations. We pose the binary hypothesis test

$$H_0 : U = U_0, \quad H_1 : U = U_1, \quad (4)$$

From (2), the conditional mean of the downstream spectrum under either hypothesis satisfies

$$\mathbb{E}[\widehat{D}(f) | H_i] = p \widehat{G}(f) \widehat{U}_i(f), \quad i \in \{0, 1\} \quad (5)$$

Hence, the frequency-wise difference of the expected downstream signals is

$$\Delta \widehat{D}(f) = p \widehat{G}(f) [\widehat{U}_0(f) - \widehat{U}_1(f)] \quad (6)$$

To quantify the detectability of this contrast, we introduce $N_{\mathcal{M}}(f) > 0$ as an observation-model-specific effective noise scale, where $\mathcal{M}$ denotes the observation model. Its precise interpretation depends on the likelihood.

For additive stationary Gaussian noise, $N_{\mathcal{M}}(f) = S_\varepsilon(f)$ is the noise spectral density, and the resulting weighted spectral distance corresponds directly to the

Gaussian likelihood geometry. For independent count observations, the variance depends on the time-varying conditional mean and does not, in general, yield a frequency-wise diagonal likelihood. In those settings, we use conservative uniform variance bounds together with Parseval's identity to obtain a spectral upper bound on the Kullback–Leibler divergence. Thus, $N_{\mathcal{M}}(f)$ should be interpreted generally as an effective likelihood-based noise scale, rather than as a literal Fourier-domain variance under every observation model.

As illustrated schematically in Fig. 1F, only the portion of upstream spectral contrast that survives delay filtering relative to the effective observation noise contributes to downstream distinguishability. We define the model-specific instantiation of the common separability functional as

$$\begin{aligned} J_{\mathcal{M}}(U_0, U_1) &= \int_{-\infty}^{\infty} \frac{|\Delta \hat{D}(f)|^2}{N_{\mathcal{M}}(f)} df \\ &= \int_{-\infty}^{\infty} \frac{|p \hat{G}(f)|^2}{N_{\mathcal{M}}(f)} \left| \hat{U}_1(f) - \hat{U}_0(f) \right|^2 df. \end{aligned} \quad (7)$$

Because $U(t)$, $g(t)$, and the downstream mean are real-valued, their Fourier transforms are conjugate symmetric, and the integrand in (7) is even in frequency. Accordingly, numerical calculations use the equivalent one-sided representation over nonnegative frequencies, with the standard factor-of-two correction for interior positive-frequency components; full discrete implementation details are provided in Appendix S1.

When the observation model is clear from context, we write $J(U_0, U_1)$ for $J_{\mathcal{M}}(U_0, U_1)$. In every case,

$$J_{\mathcal{M}}(U_0, U_1) = \int_{-\infty}^{\infty} w_{\mathcal{M}}(f) \left| \hat{U}_1(f) - \hat{U}_0(f) \right|^2 df,$$

where

$$w_{\mathcal{M}}(f) = \frac{|p \hat{G}(f)|^2}{N_{\mathcal{M}}(f)}.$$

The delay kernel therefore determines the common spectral geometry of the problem, whereas the observation model determines the scale against which the surviving downstream contrast is evaluated.

For epidemiologically realistic delay distributions, $|\hat{G}(f)|^2$ often decays rapidly with frequency. Unless the effective noise scale decreases sufficiently rapidly to offset this attenuation, the spectral weight $w_{\mathcal{M}}(f)$ becomes small at high frequencies. Consequently, differences concentrated at short temporal scales may contribute little to downstream separability even when their upstream amplitudes are substantial.

Detectability Under Different Observation Models

In the hypothesis-testing framework introduced above, the effective noise scale $N(f)$ depends on the observation model. We therefore instantiate the separability framework for three commonly used observation models: additive Gaussian noise, Poisson counting noise, and negative-binomial counting noise. Although these models induce different variance structures, they all lead to likelihood-based bounds relating the separability functional $J(U_0, U_1)$ to statistical distinguishability. Full derivations are provided in Appendix S3.

For any decision rule ϕ , define the total testing error

$$R(\phi) = P_0(\phi = H_1) + P_1(\phi = H_0),$$

where P_0 and P_1 denote probabilities under the two hypotheses defined above. Small values of $R(\phi)$ indicate that the two upstream trajectories can be reliably distinguished from downstream observations.

Likelihood-based arguments then imply that, for any target total testing error level $\alpha \in (0, 1)$, each observation model admits a corresponding non-identifiability threshold. When $J(U_0, U_1)$ falls below this threshold, every statistical test satisfies

$$\inf_{\phi} R(\phi) \geq \alpha,$$

so U_0 and U_1 cannot be reliably distinguished from downstream observations alone.

The resulting observation-model-specific quantities are summarized in Table 1.

Across all three observation models, the same qualitative mechanism emerges. Delay attenuation enters through $|\hat{G}(f)|^2$, conversion loss through p^2 , and observational uncertainty through the effective noise scale $N(f)$. Statistical distinguishability is therefore governed by the same weighted spectral distance $J(U_0, U_1)$, with observation models differing only in their variance structure and corresponding non-identifiability threshold.

Results

Empirical delays impose strong low-pass filtering

Figure 1 illustrates how delayed observation can fundamentally limit the temporal resolution of epidemic surveillance. We now ask whether this theoretical mechanism holds across published epidemiological delay distributions.

We evaluated the theoretical predictions using several published epidemiological delay distributions spanning

Table 1: **Observation-model-specific normalizations and non-identifiability thresholds.** For a target total testing error level $\alpha \in (0, 1)$, non-identifiability is guaranteed whenever $J(U_0, U_1)$ falls below the corresponding threshold. For count models, $m_i(t) = p(U_i * g)(t)$ and $0 < m_{\min} \leq m_i(t) \leq m_{\max}$. For the negative-binomial model, $V(m) = m + m^2/\kappa$ and $V_{\min} = V(m_{\min})$.

Observation model	Variance structure	Normalization $N(f)$	Non-identifiability threshold
Gaussian	$S_\varepsilon(f)$	$S_\varepsilon(f)$	$4(1 - \alpha)^2$
Poisson	$m(t)$	$m_{\min}$	$4(1 - \alpha)^2$
Negative binomial	$V(m) = m + \frac{m^2}{\kappa}$	$V_{\min}$	$4(1 - \alpha)^2$

both biological delays (exposure-to-onset) and surveillance delays (symptom-onset-to-report) (Fig. 2A,D). Together, these representative delays span diverse diseases, surveillance settings, geographic regions, and study periods, providing a broad test of whether the theoretical predictions hold across realistic epidemiological delay processes.

For each delay distribution $g(\tau)$, we computed its frequency response $\hat{G}(f)$ and attenuation profile $|\hat{G}(f)|^2$. Despite their heterogeneous epidemiological origins, all representative delay distributions exhibited rapidly decaying frequency responses (Fig. 2B,E), confirming that realistic epidemiological delays universally behave as low-pass filters.

Frequencies corresponding to weekly and sub-weekly variation (i.e., $f \geq 1/7$ cycles/day) fall in a strongly attenuated regime for many of the empirical delays. In some cases, especially for broader or more dispersed delay distributions, the power transmitted at weekly and faster frequencies is reduced by orders of magnitude relative to low-frequency components.

We next quantify how this attenuation translates into statistical detectability. The observed attenuation translates directly into a frequency-dependent detectability limit. Consider an upstream perturbation concentrated around a characteristic period T , corresponding to frequency $f_0 = 1/T$. Under the separability criterion derived above, the contribution of this perturbation to downstream distinguishability is proportional to

$$J(f_0) \propto p^2 |\hat{G}(f_0)|^2 \frac{|\Delta \hat{U}(f_0)|^2}{N(f_0)},$$

where $N(f_0)$ is the effective noise level at frequency f_0 . The quantity

$$\frac{|\Delta \hat{U}(f_0)|^2}{N(f_0)}$$

therefore represents the frequency-domain contrast-to-noise ratio, namely the upstream spectral contrast normalized by the effective downstream noise level.

For a fixed target testing-error level α , the model-specific non-identifiability threshold derived in the Methods implies that the minimum required contrast-to-noise ratio is

$$\left(\frac{|\Delta \hat{U}(f_0)|^2}{N(f_0)} \right)_{\min} = \frac{\tau_{M,\alpha}}{p^2 |\hat{G}(f_0)|^2},$$

where M denotes the observation model and $\tau_{M,\alpha}$ is the corresponding testing-error threshold. Throughout Fig. 2C,F, we illustrate this threshold under the negative-binomial observation model, for which $\tau_{M,\alpha} = 4(1 - \alpha)^2$.

Figure 2C,F visualize the minimum required frequency-domain contrast-to-noise ratio as a function of perturbation period T . The required upstream contrast increases sharply as perturbation periods become shorter, reflecting the progressively stronger attenuation of high-frequency components by the delay kernel. Lower conversion rates further amplify this effect: reducing p from 1.0 to 0.2 increases the detectability threshold by a factor of $1/p^2 = 25$. Partial ascertainment therefore compounds the information loss caused by delay filtering.

Together, these results show that empirical epidemic delay distributions impose a finite temporal resolution limit on upstream inference. The key consequence is not simply that delays smooth the observed data, but that they make short-timescale upstream variation statistically difficult to distinguish from noise. Perturbations occurring over several weeks may remain visible after delay convolution, whereas weekly or sub-weekly fluctuations often require unrealistically large contrast-to-noise ratios to be identifiable.

Spectral identifiability constrains epidemiological inference

The practical consequence of non-identifiability is not simply that multiple upstream trajectories can explain the same downstream observations. Rather, whenever downstream data fail to identify part of the upstream process, reconstruction necessarily relies on information ex-

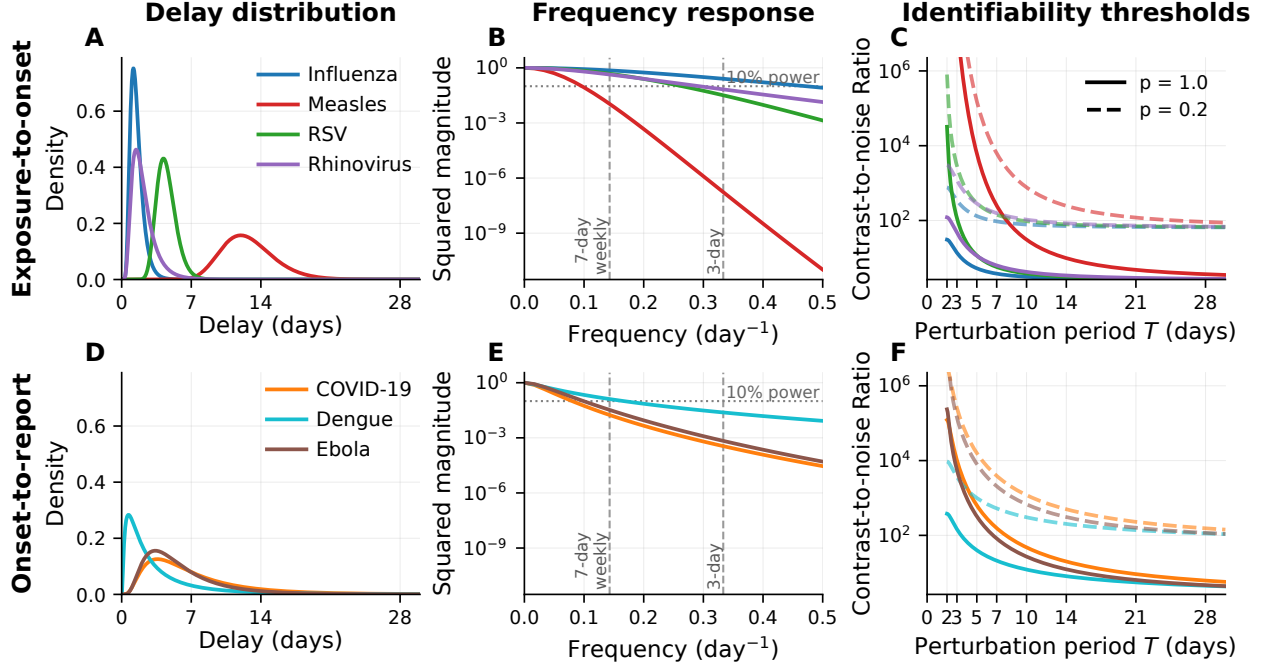


Figure 2: **Empirical epidemic delays impose a frequency-dependent limit on upstream identifiability.** (A,D) Representative published delay distributions for exposure-to-onset ([25]) and onset-to-report processes across multiple diseases ([32], [3], [13], [19]). (B,E) Corresponding squared magnitude $|\hat{G}(f)|^2$ of the delay kernels. Vertical dashed lines indicate frequencies corresponding to 7-day and 3-day periods. The dotted horizontal line marks the 10% power level. (C,F) Minimum required frequency-domain contrast-to-noise ratio needed for statistical distinguishability as a function of perturbation period $T = 1/f$, derived from the separability criterion. Solid and dashed curves correspond to conversion rate $p = 1.0$ and $p = 0.2$, respectively.

ternal to the data, including prior distributions, smoothness assumptions, regularization constraints, or mechanistic model structure [22, 30, 28]. These assumptions play an essential role in stabilizing ill-posed inverse problems. However, they also imply that weakly identified components of a reconstructed trajectory are determined jointly by the observed data and by the assumptions used to complete the inference problem. Our objective is therefore not simply to characterize uncertainty in reconstructed incidence curves, but to determine which epidemiological conclusions are actually supported by the information preserved in delayed and noisy observations.

To illustrate this mechanism without comparing specific reconstruction algorithms, we constructed frequency-constrained upstream trajectories from the same downstream COVID-19 incidence series. For each cutoff frequency f_c , we estimated an upstream trajectory $U_{f_c}^*$ restricted to Fourier components satisfying $f \leq f_c$, while requiring its delayed projection $KU_{f_c}^*$ to remain consistent with the same downstream observations (Methods; Appendix S4). We considered $f_c = 1/35$, $1/14$, and $1/7$ cycles/day, corresponding to characteristic periods of 35, 14, and 7 days. These trajectories are not intended

as alternative epidemiological models. Instead, they illustrate how progressively admitting higher-frequency structure changes the range of admissible reconstructions.

To visualize this uncertainty, we computed frequency-constrained non-identifiable perturbation bands around each reference trajectory. These bands consist of perturbations whose downstream effect remains below the non-identifiability threshold implied by the separability criterion J (Appendix S4). The resulting bands widened as progressively higher-frequency components were admitted, illustrating that the set of upstream trajectories compatible with the same downstream observations expands when weaker frequency constraints are relaxed (Fig. 3A).

In practice, reconstructed incidence curves are rarely the final inferential target. Instead, they are commonly used to derive epidemiological quantities such as epidemic peak timing, peak magnitude, and short-term growth rates. To examine how spectral identifiability translates into epidemiological inference, we evaluated these three representative quantities across the admissible reconstruction family. For visualization, peak timing was expressed as the normalized location of the epidemic peak within the study period, and peak magnitude was normal-

ized by the largest reconstructed peak across the admissible reconstruction family (Fig. 3D).

Although all admissible reconstructions generated nearly identical downstream trajectories (Fig. 3B), they did not support the same epidemiological conclusions. Epidemic peak timing and peak magnitude remained remarkably consistent across the reconstruction family (Fig. 3D), whereas inferred seven-day growth-rate trajectories differed substantially (Fig. 3C).

To understand this contrast, consider the definition of the seven-day log growth rate,

$$r_7(t) = \frac{1}{7} [\log\{U(t+7) + c\} - \log\{U(t) + c\}], \quad (8)$$

where $c > 0$ is a small numerical offset.

Unlike peak timing or peak magnitude, the seven-day growth rate is computed from local temporal differences. Temporal differencing emphasizes short-timescale variation (Appendix S4), causing growth-rate estimates to depend disproportionately on high-frequency components of the reconstructed trajectory. Under delayed observation, these are precisely the components that are most strongly attenuated by the delay kernel and least constrained by the downstream observations.

Consequently, the differences in the inferred seven-day growth-rate trajectories (Fig. 3C) cannot be attributed to the downstream observations alone. Rather, they arise from alternative completions of weakly identified high-frequency components that remain effectively unconstrained by the observed data. In this sense, short-term growth-rate estimates are determined jointly by the information contained in the observations and by the assumptions used to reconstruct the weakly identified portion of the upstream trajectory.

These results demonstrate that agreement in downstream goodness-of-fit does not imply equal support for all epidemiological conclusions. Reconstructions that are statistically indistinguishable at the observation level may nevertheless lead to different scientific inferences. Spectral identifiability therefore provides a principled way to distinguish quantities that are constrained by the information contained in delayed observations from those that depend primarily on assumptions introduced to resolve the weakly identified components of the inverse problem.

Event impacts form reference-conditioned low-separability landscapes

Epidemiological events are often known independently of surveillance data. Mass gatherings are scheduled, temporary lockdowns are announced, and vaccination campaigns are documented administratively. The principal inferential question is therefore often not whether an event

occurred, but which transmission impacts are distinguishable from competing explanations after delay filtering and observational noise. In particular, knowledge of an event's timing does not imply that its magnitude or temporal footprint can be uniquely recovered from downstream surveillance observations.

We characterized this uncertainty using reference-conditioned event-identifiability landscapes. For a specified reference event profile, these landscapes map alternative combinations of event duration and transmission impact according to their downstream separability after epidemic propagation, delay convolution, and observation noise. We considered three stylized event families representing common epidemiological perturbations: a transient increase in transmission associated with a mass gathering, a transient reduction associated with a temporary lockdown, and a gradual reduction associated with a vaccination rollout. The reference profiles corresponded to a 50% peak proportional change in R_t , with characteristic durations of 3, 10, and 60 days, respectively.

We represented each event through a transmission multiplier $m(t; \theta)$ acting on a baseline reproduction-number trajectory,

$$R(t; \theta) = R_0(t)m(t; \theta), \quad (9)$$

where θ specifies the event duration (or rollout timescale) and the peak proportional change in R_t . The three transmission multipliers are summarized in Table 2.

To examine how alternative event profiles propagate through epidemic dynamics and delayed observation, we conducted a controlled synthetic experiment. Synthetic epidemics provide a common baseline for comparison while avoiding the confounding effects of multiple overlapping drivers present in real outbreaks. Across all event classes, we used the same baseline epidemic trajectory, generation-interval distribution, renewal framework, delay kernel, conversion rate, and negative-binomial observation model with a common dispersion parameter. Within each event family, every candidate profile was generated by perturbing the same baseline epidemic through the same renewal process, so that differences in J_{event} reflected only differences in the event profile under a common epidemic setting. Complete simulation details are provided in Appendix S5.

Incidence was generated through the renewal equation,

$$U(t; \theta) = R(t; \theta) \sum_k w_k U(t - k; \theta), \quad (10)$$

and mapped to the downstream mean through the same delay kernel used throughout the analysis, where w_k denotes the generation-interval distribution.

For a specified reference event θ^* , we compared each

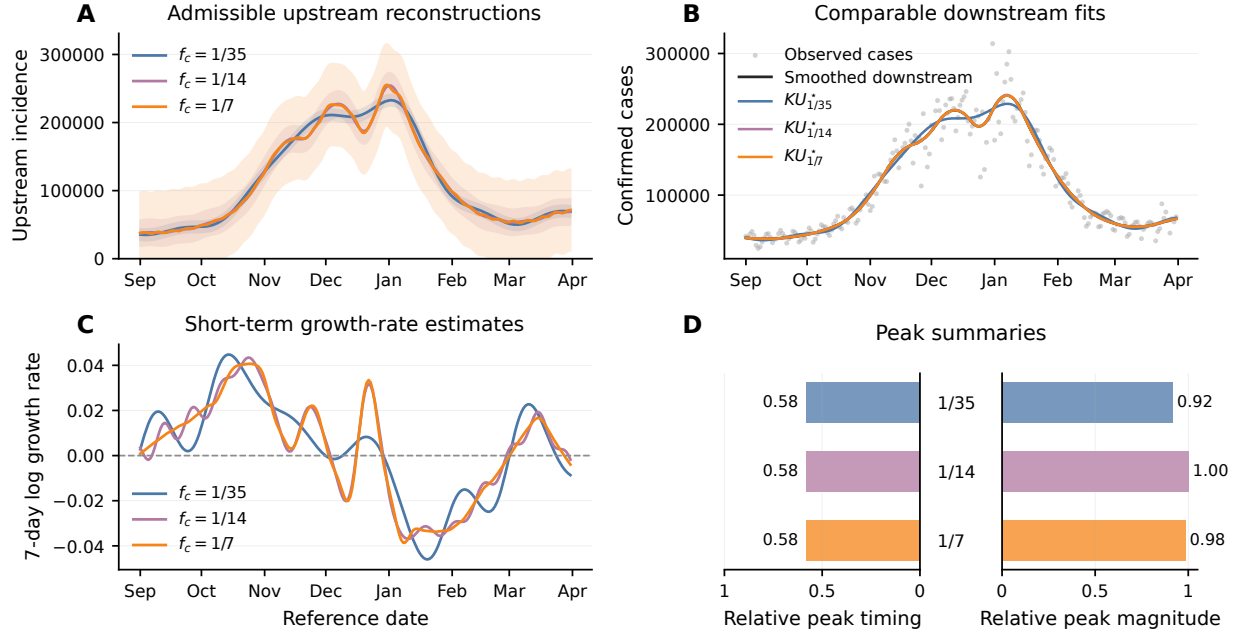


Figure 3: **Spectral identifiability constrains epidemiological inference.** Frequency-constrained upstream reconstructions with cutoff frequencies $f_c = 1/35$, $1/14$, and $1/7$ cycles/day. (A) Admissible upstream reconstructions with corresponding non-identifiable perturbation bands. (B) Delayed projections of the reconstructed trajectories together with the observed and smoothed downstream case series. (C) Seven-day log growth rates derived from the reconstructed upstream trajectories. (D) Relative peak timing and relative peak magnitude of the admissible reconstructions.

candidate profile θ through the induced upstream contrast

$$\Delta U(t; \theta, \theta^*) = U(t; \theta) - U(t; \theta^*), \quad (11)$$

and quantified its downstream separability using the spectral functional

$$J_{\text{event}}(\theta, \theta^*) = \int \frac{|p\hat{G}(f)|^2 |\Delta\hat{U}(f; \theta, \theta^*)|^2}{N(f)} df. \quad (12)$$

We set $p = 1$ in the controlled examples. The resulting landscape is necessarily reference-conditioned: changing θ^* changes the position and geometry of its low-separability neighborhood, but not the procedure used to construct it.

The resulting landscapes reveal extended low-separability neighborhoods around each reference event rather than isolated identifiable parameter values (Fig. 4A,F,K). Within the cyan contour, the testing-error bound derived in Methods guarantees that every statistical decision rule has total error at least α under the specified observation model. Candidate profiles in this region therefore cannot be reliably distinguished from the reference at the chosen error criterion. Beyond this boundary, separability increases continuously with

J_{event} , but distinguishability is not guaranteed because the bound is sufficient rather than necessary.

The resulting landscapes reveal event-specific trade-offs between transmission magnitude and event duration. For transient mass gatherings and temporary lockdowns, shorter but stronger perturbations can remain only weakly separated from longer but weaker perturbations. By contrast, vaccination rollout generates a more localized landscape because changes in rollout duration primarily alter lower-frequency epidemic dynamics. These geometries are not intrinsic properties of the event labels themselves; they emerge from the interaction among epidemic dynamics, delay filtering, and the spectral content of the induced transmission perturbation.

To illustrate the practical implications of these landscapes, we selected one representative alternative for each event class lying immediately outside the sufficient non-identifiability boundary while maximizing upstream trajectory contrast (Appendix S5). As shown in Fig. 4B,C,G,H,L,M, the reference and alternative profiles produce visibly different transmission multipliers and upstream incidence trajectories. After delay convolution, however, much of this contrast is attenuated, and the resulting downstream means remain similar relative to observational variability. A single synthetic realization gen-

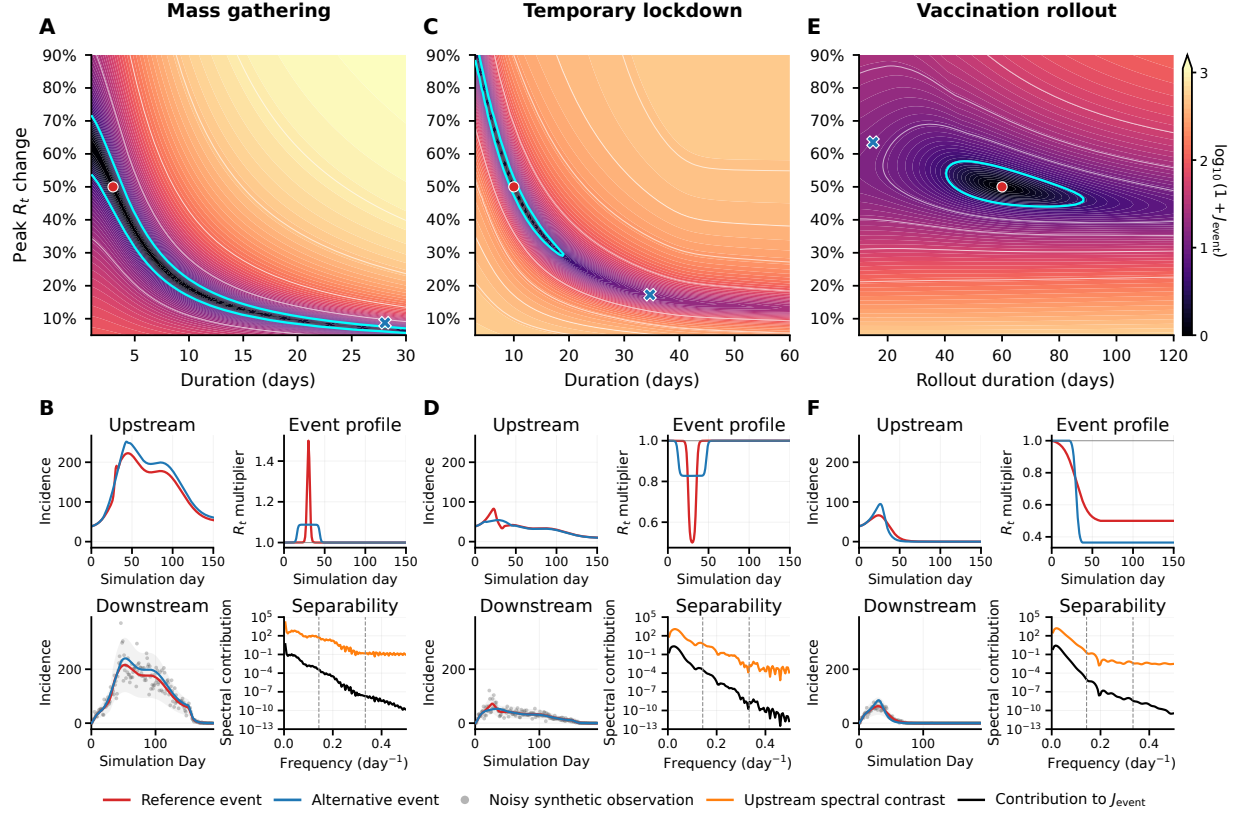


Figure 4: Event impacts occupy extended regions of weak downstream separability. Each column presents a reference-conditioned event-impact analysis for a mass gathering, temporary lockdown, or vaccination rollout. (A,C,E) Event-identifiability landscapes over duration, or rollout duration, and peak proportional change in R_t . Color denotes $\log_{10}(1 + J_{\text{event}})$, with darker regions indicating weaker downstream separation from the reference. Red circles mark the reference profiles. Cyan contours denote the sufficient non-identifiability boundary $J_{\text{event}} = 4(1 - \alpha)^2$, with $\alpha = 0.20$, under the negative-binomial observation model and effective noise scale defined in Methods. (B,D,F) Propagation of each reference-alternative contrast through the inferential pipeline. Red and blue curves denote the reference and alternative profiles, respectively. Top left: upstream incidence. Top right: event-specific transmission multiplier. Bottom left: synthetic delayed downstream means and one noisy realization generated under the reference event, shown as gray points. Bottom right: upstream spectral contrast $|\Delta\hat{U}(f)|^2$ and its delay- and noise-weighted contribution to downstream separability, $|p\hat{G}(f)|^2 |\Delta\hat{U}(f)|^2 / N(f)$. Vertical dashed lines indicate frequencies corresponding to 7-day and 3-day periods. The zero-frequency component is included in J_{event} but omitted from the displayed spectra.

Table 2: **Stylized event models used in the event-impact analysis.** The parameter a denotes the peak proportional change in R_t , and τ controls the event duration or rollout timescale.

Event type	Transmission multiplier	Reference profile
Mass gathering	$1 + a h_\tau(t - t_0)$	50% increase, 3-day duration
Temporary lockdown	$1 - a h_\tau(t - t_0)$	50% reduction, 10-day duration
Vaccination rollout	$1 - a C_\tau(t)$	50% reduction, 60-day rollout

The event window $h_\tau(t - t_0)$ is normalized to attain a maximum of one and is localized around the known event time t_0 . The rollout curve $C_\tau(t)$ increases monotonically from zero to one over a timescale controlled by τ .

erated under the reference profile is visually compatible with both alternatives, illustrating that substantial ambiguity can persist even beyond the conservative theoretical boundary.

The spectral decomposition identifies the source of this ambiguity. At each frequency, the upstream contrast contributes to separability in proportion to

$$\frac{|p\hat{G}(f)|^2}{N(f)} \left| \Delta\hat{U}(f) \right|^2.$$

The upstream contrast spectrum and its statistically usable downstream contribution can therefore differ sharply. Rapidly varying components are strongly compressed by delay attenuation and observational noise, whereas only contrasts overlapping sufficiently informative frequencies contribute materially to J_{event} . A substantial difference in upstream incidence consequently need not produce a correspondingly large amount of distinguishable downstream information.

Together, these results distinguish event occurrence from event-impact identifiability. Even when the timing and existence of an epidemiological event are known, delayed surveillance data may support a neighborhood of epidemiologically distinct transmission impacts rather than a unique estimate of event magnitude and duration. The event-identifiability landscape therefore provides a quantitative characterization of the competing transmission histories that remain only weakly separated from a specified reference under a given epidemic and observation model.

Conclusion

Reporting delays are commonly understood as shifting and smoothing epidemic signals. Our results show that their more consequential effect is to impose a quantitative, frequency-dependent limit on what can be inferred from downstream surveillance data. By combining the transfer function of the delay process with conversion and observation-model-specific uncertainty, we developed a spectral separability framework that characterizes whether competing upstream epidemic trajectories

can be statistically distinguished after they pass through the observation process. This perspective reframes epidemic reconstruction from recovering a latent trajectory to determining which differences between epidemic histories, and which scientific conclusions derived from them, remain supported by the information transmitted downstream.

Across empirical delay distributions, this framework revealed a consistent hierarchy across temporal scales and inferential targets. Slow epidemic variation is comparatively well preserved, whereas short-term fluctuations e.g. weekly and sub-weekly fluctuations can be attenuated by orders of magnitude and may require unrealistically large upstream contrasts to remain distinguishable from noise. This loss of information does not affect all derived quantities equally. Features governed primarily by broad, low-frequency structure, such as the timing of a major epidemic wave, can remain comparatively stable across admissible reconstructions, while quantities based on local temporal differences, such as short-term growth rates, depend more strongly on weakly identified high-frequency components. The same principle extends to event attribution. Even when the occurrence and timing of a lockdown, mass gathering, or vaccination campaign are known, its transmission magnitude and temporal footprint may occupy an extended low-separability region rather than correspond to a uniquely identifiable parameterization. Identifiability is therefore not a binary property of an entire epidemic trajectory. It is specific to the temporal scale, epidemiological feature, and scientific comparison being considered.

These limits clarify the role of regularization, prior distributions, and mechanistic structure in epidemic reconstruction. Such assumptions are indispensable for stabilizing an ill-posed inverse problem and for producing epidemiologically coherent estimates. They cannot, however, recreate information that has been strongly suppressed by the observation process. When the likelihood provides little discrimination among alternative high-frequency trajectories, different priors, smoothing penalties, process models, or preprocessing choices may select different solutions while achieving nearly indistin-

guishable downstream fits. A reconstruction may consequently be numerically stable, have narrow conditional uncertainty, and closely reproduce the observed data without its fine-scale features being strongly identified by those data. Downstream goodness of fit, inferential precision, and statistical identification should therefore be treated as distinct properties.

The quantitative boundary between identifiable and weakly identifiable variation is conditional on the observation process through which epidemic information is transmitted. Among its components, uncertainty in the delay distribution is particularly consequential because the delay kernel directly determines the frequency-dependent attenuation of upstream variation. Our sensitivity analysis shows that moderate changes in the assumed median delay or dispersion can shift the identifiability boundary by orders of magnitude for rapid perturbations, whereas longer-timescale variation is generally less sensitive to the same uncertainty (Appendix S6; Fig. 5). Consequently, the temporal resolution limit should be evaluated across epidemiologically plausible delay specifications rather than under a single fixed kernel, particularly when inference concerns temporal scales near the resolution boundary.

Other components of the observation process—including ascertainment, observation noise, and the observation window—likewise influence the quantitative location of this boundary. Complementary observations, such as wastewater measurements, testing data, mobility or contact indicators, genomic information, and administrative records of interventions, can further reduce ambiguity by contributing information unavailable from a single delayed outcome stream. Their contribution should be interpreted as providing additional sources of identification, rather than as evidence that the original surveillance series itself contained the missing temporal information.

These findings have implications for both surveillance design and the evaluation of epidemic models. Shortening and narrowing reporting delays, increasing ascertainment, reducing measurement noise, and combining observations with complementary temporal responses do more than improve timeliness: they increase the range of epidemic variation that can be statistically resolved. Model evaluation should ask not only whether an estimate fits observed outcomes or produces a precise posterior, but also whether the particular epidemiological quantity being reported is identifiable under the assumed observation process and remains stable across plausible delay specifications and reconstruction assumptions. Making these limits explicit would shift epidemic inference away from presenting a single detailed latent history toward reporting which temporal scales, epidemiological features, and scientific conclusions are actually supported by the available

surveillance system.

Data Availability

The code used to generate all figures and analyses is publicly available at <https://github.com/jingjtang/spectral-identifiability>.

Acknowledgments

We thank Valerie Ventura, Logan Brooks, Will Townes, and Ryan Tibshirani for several enlightening conversations.

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S1 Construction of the illustrative example in Figure 1

Figure 1 provides a data-informed illustration of how delay filtering suppresses rapid differences between candidate upstream epidemic trajectories. The figure is intended to illustrate the observation and separability framework rather than to provide a substantive reconstruction of the U.S. epidemic.

S1.1 Observed downstream incidence

Daily U.S. confirmed-case incidence was obtained from the Johns Hopkins University Center for Systems Science and Engineering COVID-19 data stream [10], accessed through the Delphi Epidata API [14]. We used the national-level `jhu-csse/confirmed_incidence_num` signal from April 1, 2020, through June 30, 2021. The series was aligned to a complete daily sequence, with missing values linearly interpolated. Figure 1

displays the period from September 1, 2020, through March 31, 2021.

S1.2 Smoothed downstream mean and observation noise

To separate broad epidemic variation from day-to-day observational fluctuation, we estimated a smooth downstream mean $\hat{\mu}(t)$ using second-order trend filtering. The effective degrees of freedom were selected over a pre-specified grid. Conditional on each candidate smooth mean, Gaussian, Poisson, and negative-binomial observation models were fitted, and the combination of trend-filter complexity and observation model was selected by the Akaike information criterion. The resulting residual or model-based variance scale was used to define the effective observation-noise level shown in Fig. 1F.

S1.3 Delay distribution

The delay kernel was based on the published COVID-19 symptom-onset-to-case-report distribution reported by Zhang et al. [32]. A lognormal representation was converted to a discrete daily probability mass function by integrating the fitted distribution over consecutive one-day intervals. The kernel was truncated at 60 days and renormalized to sum to one.

S1.4 Construction of the reference upstream trajectory

The reference upstream trajectory $U_0(t)$ was obtained by solving a nonnegative regularized inverse problem. Let

K_g denote the linear convolution operator induced by the delay kernel g . We estimated U_1 by minimizing

$$\|K_g U - \hat{\mu}\|_2^2 + \lambda \|\Delta^2 U\|_1$$

subject to $U(t) \geq 0$, where Δ^2 is the second-difference operator. The regularization term stabilizes the inverse problem and produces a smooth reference trajectory whose delayed projection follows the estimated downstream mean.

S1.5 Construction of the alternative upstream trajectory

The alternative trajectory $U_1(t)$ was constructed to differ visibly from $U_0(t)$ at short temporal scales while producing a similar downstream projection. Candidate perturbations were formed from mixtures of sinusoidal components concentrated near frequencies corresponding to 7- and 3-day periods. For each candidate perturbation $h(t)$, we considered

$$U_1(t) = \max\{U_0(t) + ah(t), 0\},$$

where the amplitude a was increased to the largest value satisfying both

$$\text{RMSE}(K_g U_1, K_g U_0) \leq \gamma \text{RMSE}(K_g U_0, y)$$

and

$$\text{RMSE}(K_g U_1, y) \leq \eta \text{RMSE}(K_g U_0, y).$$

For the displayed example, $\gamma = 1$ and $\eta = 1.05$. Among feasible candidates, we selected the perturbation producing the largest visible upstream separation while retaining substantial spectral energy near the two target frequencies.

S1.6 Spectral quantities

The upstream spectral contrast shown in Fig. 1D is

$$\left| \hat{U}_0(f) - \hat{U}_1(f) \right|^2.$$

The delay attenuation in Fig. 1E is

$$\left| \hat{G}(f) \right|^2,$$

and the filtered contrast in Fig. 1F is

$$p^2 \left| \hat{G}(f) \right|^2 \left| \hat{U}_0(f) - \hat{U}_1(f) \right|^2.$$

The horizontal reference line denotes the effective observation-noise level under the selected observation model. These quantities are shown to illustrate the frequency-wise components that motivate the separability functional introduced in the main text.

S1.7 Discrete one-sided evaluation of the separability functional

The separability functional in Eq. (7) is defined using a two-sided continuous Fourier representation. In numerical calculations, however, the upstream trajectories and delay kernels are real-valued, discretely sampled sequences. Their discrete Fourier transforms therefore satisfy conjugate symmetry, allowing the equivalent calculation to be performed using only the nonnegative-frequency coefficients.

Let

$$\Delta U_n = U_{1,n} - U_{0,n}, \quad n = 0, \dots, n_U - 1,$$

denote the discrete upstream contrast, and let

$$g_n, \quad n = 0, \dots, n_g - 1,$$

denote the discrete delay probability mass function. The corresponding difference between the expected downstream trajectories is the linear convolution

$$\Delta D_n = p(\Delta U * g)_n.$$

To represent this linear convolution using the discrete Fourier transform, both sequences are zero-padded to the full convolution length

$$L = n_U + n_g - 1. \quad (13)$$

This padding prevents the circular wraparound that would otherwise arise from multiplication in the discrete Fourier domain.

For a sampling interval Δt , define the length- L discrete Fourier transforms

$$\Delta \tilde{U}_m = \sum_{n=0}^{L-1} \Delta U_n \exp\left(-\frac{2\pi i m n}{L}\right), \quad (14)$$

$$\tilde{G}_m = \sum_{n=0}^{L-1} g_n \exp\left(-\frac{2\pi i m n}{L}\right), \quad (15)$$

evaluated at frequencies

$$f_m = \frac{m}{L\Delta t}. \quad (16)$$

Under this convention, the Fourier transform of the downstream contrast is

$$\Delta \tilde{D}_m = p \tilde{G}_m \Delta \tilde{U}_m. \quad (17)$$

Because ΔU_n and g_n are real-valued,

$$\Delta \tilde{U}_{L-m} = \Delta \tilde{U}_m^*, \quad \tilde{G}_{L-m} = \tilde{G}_m^*, \quad (18)$$

where $*$ denotes complex conjugation. Consequently, the spectral integrand is symmetric between positive and negative frequencies. The two-sided spectral sum can therefore be evaluated from the one-sided real Fourier transform over

$$m = 0, \dots, \left\lfloor \frac{L}{2} \right\rfloor.$$

The discrete one-sided approximation to the model-specific separability functional is

$$J_{\mathcal{M}}^{(L)}(U_0, U_1) = \frac{\Delta t}{L} \sum_{m=0}^{\lfloor L/2 \rfloor} q_m \frac{|p \tilde{G}_m|^2}{N_{\mathcal{M}}(f_m)} |\Delta \tilde{U}_m|^2, \quad (19)$$

where the one-sided multiplicity weights are

$$q_m = \begin{cases} 1, & m = 0, \\ 1, & L \text{ is even and } m = L/2, \\ 2, & \text{otherwise.} \end{cases} \quad (20)$$

The zero-frequency coefficient receives unit weight because it has no distinct negative-frequency counterpart. When L is even, the Nyquist coefficient $m = L/2$ is also self-conjugate and therefore receives unit weight. Every other positive-frequency coefficient represents a conjugate pair and consequently receives weight two.

The normalization in Eq. (19) follows from Parseval's identity under the discrete Fourier-transform convention above:

$$\Delta t \sum_{n=0}^{L-1} |x_n|^2 = \frac{\Delta t}{L} \sum_{m=0}^{L-1} |\tilde{X}_m|^2. \quad (21)$$

Thus, the one-sided weighted sum in Eq. (19) recovers exactly the same spectral energy as the corresponding two-sided discrete sum.

For daily observations, $\Delta t = 1$ day, and Eq. (19) simplifies to

$$J_{\mathcal{M}}^{(L)}(U_0, U_1) = \frac{1}{L} \sum_{m=0}^{\lfloor L/2 \rfloor} q_m \frac{|p \tilde{G}_m|^2}{N_{\mathcal{M}}(f_m)} |\Delta \tilde{U}_m|^2. \quad (22)$$

For additive Gaussian noise with a constant standard deviation σ_ε , the effective noise scale is

$$N_{\mathcal{M}}(f_m) = \sigma_\varepsilon^2,$$

giving

$$J_G^{(L)}(U_0, U_1) = \frac{1}{L} \sum_{m=0}^{\lfloor L/2 \rfloor} q_m \frac{|p \tilde{G}_m|^2 |\Delta \tilde{U}_m|^2}{\sigma_\varepsilon^2}. \quad (23)$$

More generally, a frequency-dependent effective noise scale $N_{\mathcal{M}}(f_m)$ can be inserted directly into Eq. (19). For

the count observation models considered in this study, the corresponding conservative likelihood-based variance normalization is constant across frequency, as specified in Table 1 and derived in Appendix S1.

Equation (19) is therefore the discrete numerical counterpart of the continuous two-sided functional in Eq. (7). The use of a one-sided real Fourier transform changes only the computational representation and does not change the value or interpretation of the separability functional.

S3 Spectral separability bounds under common observation models

Throughout this appendix, we consider the simple hypotheses

$$H_0 : U = U_0, \quad H_1 : U = U_1,$$

and let P_0 and P_1 denote the corresponding laws of the downstream observations.

For any decision rule ϕ , define the total testing error

$$R(\phi) = P_0(\phi = H_1) + P_1(\phi = H_0).$$

For two simple hypotheses,

$$\inf_{\phi} R(\phi) = 1 - \|P_0 - P_1\|_{TV}.$$

Pinsker's inequality therefore gives

$$\inf_{\phi} R(\phi) \geq 1 - \sqrt{\frac{1}{2} D_{KL}(P_0 \| P_1)}. \quad (24)$$

We use the common spectral separability functional

$$J(U_0, U_1) = \int_{-\infty}^{\infty} \frac{|p\hat{G}(f)|^2}{N(f)} \left| \hat{U}_0(f) - \hat{U}_1(f) \right|^2 df, \quad (25)$$

where the normalization $N(f)$ is specified by the observation model below.

S3.1 Gaussian observation noise

S3.1.1 Setup and spectral representation

Suppose that

$$D(t) = p(U * g)(t) + \varepsilon(t),$$

where $\varepsilon(t)$ is a mean-zero stationary Gaussian process with spectral density $S_{\varepsilon}(f) > 0$.

We use the Fourier-domain Gaussian sequence representation in which the covariance operator is diagonal in the Fourier basis. Under this representation,

$$\hat{\varepsilon}(f) \sim \mathcal{CN}(0, S_{\varepsilon}(f)),$$

with independent coefficients across frequencies. Consequently, under H_i ,

$$\hat{D}(f) \mid H_i \sim \mathcal{CN}(p\hat{G}(f)\hat{U}_i(f), S_{\varepsilon}(f)), \quad i \in \{1, 2\}.$$

S3.1.2 KL divergence and separability

Because the two Gaussian laws have the same covariance and differ only in their means, their Kullback–Leibler divergence is

$$D_{KL}(P_0 \| P_1) = \frac{1}{2} \int_{-\infty}^{\infty} \frac{|p\hat{G}(f)|^2}{S_{\varepsilon}(f)} \left| \hat{U}_0(f) - \hat{U}_1(f) \right|^2 df.$$

Thus, with

$$N(f) = S_{\varepsilon}(f),$$

equation (25) satisfies

$$D_{KL}(P_0 \| P_1) = \frac{1}{2} J(U_0, U_1). \quad (26)$$

Substituting equation (26) into equation (24) gives

$$\inf_{\phi} R(\phi) \geq 1 - \sqrt{\frac{J(U_0, U_1)}{4}}.$$

Therefore, for any target total testing error $\alpha \in (0, 1)$, if

$$J(U_0, U_1) \leq 4(1 - \alpha)^2, \quad (27)$$

then every test satisfies

$$R(\phi) \geq \alpha.$$

S3.2 Poisson observation noise

S3.2.1 Setup and downstream mean difference

Suppose that observations are conditionally independent across time and satisfy

$$D(t) \mid H_0 \sim \text{Pois}(m_0(t)), \quad D(t) \mid H_1 \sim \text{Pois}(m_1(t)),$$

where

$$m_0(t) = p(U_0 * g)(t), \quad m_1(t) = p(U_1 * g)(t).$$

Assume that both mean processes are uniformly bounded:

$$0 < m_{\min} \leq m_i(t) \leq m_{\max} < \infty, \quad i \in \{0, 1\}, \quad \forall t.$$

Define

$$\Delta m(t) = m_1(t) - m_0(t) = p[(U_1 - U_0) * g](t).$$

Taking Fourier transforms gives

$$\Delta \hat{m}(f) = p\hat{G}(f) [\hat{U}_1(f) - \hat{U}_0(f)].$$

Parseval's identity therefore yields

$$\sum_t \{\Delta m(t)\}^2 = \int_{-\infty}^{\infty} |p\hat{G}(f)|^2 \left| \hat{U}_0(f) - \hat{U}_1(f) \right|^2 df. \quad (28)$$

S3.2.2 KL divergence and quadratic bound

For scalar Poisson distributions,

$$d_{\text{Pois}}(\lambda_0, \lambda_1) = \lambda_0 \log \left(\frac{\lambda_0}{\lambda_1} \right) + \lambda_1 - \lambda_0.$$

Equivalently, this divergence is the Bregman divergence generated by

$$\psi_{\text{Pois}}(m) = m \log m - m,$$

whose second derivative is

$$\psi''_{\text{Pois}}(m) = \frac{1}{m}.$$

Taylor's theorem therefore gives

$$d_{\text{Pois}}(\lambda_0, \lambda_1) = \frac{(\lambda_1 - \lambda_0)^2}{2\xi}$$

for some ξ between λ_0 and λ_1 .

Since $\xi \geq m_{\min}$,

$$d_{\text{Pois}}(\lambda_0, \lambda_1) \leq \frac{(\lambda_1 - \lambda_0)^2}{2m_{\min}}.$$

Applying this bound at each time point and using conditional independence gives

$$D_{\text{KL}}(P_0 \| P_1) \leq \frac{1}{2m_{\min}} \sum_t \{\Delta m(t)\}^2. \quad (29)$$

S3.2.3 Spectral separability and testing error

For Poisson observations, set

$$N(f) = m_{\min}.$$

Equations (25) and (28) then give

$$J(U_0, U_1) = \frac{1}{m_{\min}} \sum_t \{\Delta m(t)\}^2.$$

Consequently, equation (29) becomes

$$D_{\text{KL}}(P_0 \| P_1) \leq \frac{1}{2} J(U_0, U_1). \quad (30)$$

Combining equations (24) and (30) yields

$$\inf_{\phi} R(\phi) \geq 1 - \sqrt{\frac{J(U_0, U_1)}{4}}.$$

Therefore, if

$$J(U_0, U_1) \leq 4(1 - \alpha)^2, \quad (31)$$

then every test satisfies

$$R(\phi) \geq \alpha.$$

S3.3 Negative-binomial observation noise

S3.3.1 Setup and mean–variance structure

Suppose that observations are conditionally independent across time and satisfy

$$D(t) | H_0 \sim \text{NB}(m_0(t), \kappa), \quad D(t) | H_1 \sim \text{NB}(m_1(t), \kappa),$$

where

$$m_0(t) = p(U_0 * g)(t), \quad m_1(t) = p(U_1 * g)(t),$$

and the dispersion parameter $\kappa > 0$ is shared by the two hypotheses.

Under this parameterization,

$$\mathbb{E}[D(t) | H_i] = m_i(t),$$

and

$$\text{Var}[D(t) | H_i] = V(m_i(t)),$$

where the negative-binomial variance function is

$$V(m) = m + \frac{m^2}{\kappa} = m \left(1 + \frac{m}{\kappa} \right). \quad (32)$$

Assume

$$0 < m_{\min} \leq m_i(t) \leq m_{\max} < \infty, \quad i \in \{0, 1\}, \quad \forall t.$$

Define

$$V_{\min} = V(m_{\min}) = m_{\min} + \frac{m_{\min}^2}{\kappa},$$

and

$$\Delta m(t) = m_1(t) - m_0(t) = p[(U_1 - U_0) * g](t).$$

S3.3.2 KL divergence and quadratic bound

For a negative-binomial family with fixed dispersion parameter κ , the scalar KL divergence can be represented as the Bregman divergence generated by

$$\psi_{\text{NB}}(m) = m \log m - (m + \kappa) \log(m + \kappa),$$

up to an additive constant that does not affect the divergence.

Its second derivative is

$$\psi''_{\text{NB}}(m) = \frac{1}{m} - \frac{1}{m + \kappa} = \frac{1}{m(1 + m/\kappa)} = \frac{1}{V(m)}.$$

Taylor's theorem therefore gives

$$d_{\text{NB}}(\lambda_0, \lambda_1) = \frac{(\lambda_1 - \lambda_0)^2}{2V(\xi)} \quad (33)$$

for some ξ between λ_0 and λ_1 .

Because $V(m)$ is increasing for $m > 0$, the bounded-mean assumption implies

$$V(\xi) \geq V(m_{\min}) = V_{\min}.$$

Hence,

$$d_{\text{NB}}(\lambda_0, \lambda_1) \leq \frac{(\lambda_1 - \lambda_0)^2}{2V_{\min}}.$$

Applying this bound at every time point and using conditional independence gives

$$D_{\text{KL}}(P_0 \| P_1) \leq \frac{1}{2V_{\min}} \sum_t \{\Delta m(t)\}^2. \quad (34)$$

S3.3.3 Spectral separability and testing error

Parseval's identity gives

$$\sum_t \{\Delta m(t)\}^2 = \int_{-\infty}^{\infty} |p\hat{G}(f)|^2 \left| \hat{U}_0(f) - \hat{U}_1(f) \right|^2 df. \quad (35)$$

For negative-binomial observations, set

$$N(f) = V_{\min}.$$

Equations (25) and (35) then give

$$J(U_0, U_1) = \frac{1}{V_{\min}} \sum_t \{\Delta m(t)\}^2.$$

Consequently, equation (34) becomes

$$D_{\text{KL}}(P_0 \| P_1) \leq \frac{1}{2} J(U_0, U_1). \quad (36)$$

Combining equations (24) and (36) gives

$$\inf_{\phi} R(\phi) \geq 1 - \sqrt{\frac{J(U_0, U_1)}{4}}.$$

Therefore, if

$$J(U_0, U_1) \leq 4(1 - \alpha)^2, \quad (37)$$

then every test satisfies

$$R(\phi) \geq \alpha.$$

S4 Frequency-constrained reconstruction and feature analysis

S4.1 Data and downstream target

We used daily confirmed COVID-19 case incidence from the Johns Hopkins University Center for Systems Science

and Engineering data stream, accessed through the Delphi Epidata API. The reconstruction analysis was performed on the national US series from April 1, 2020, through June 30, 2021. For visualization and calculation of the epidemiological feature summaries, we restricted the displayed interval to September 1, 2020, through March 31, 2021.

Let

$$y = (y_1, \dots, y_T)^\top$$

denote the observed daily case-incidence series. Because the raw observations contain day-of-week effects and other short-timescale reporting variation, we first estimated a smooth downstream mean

$$\hat{\mu} = (\hat{\mu}_1, \dots, \hat{\mu}_T)^\top$$

using second-order trend filtering. Specifically, for each target effective degrees of freedom in

$$\mathcal{D} = \{10, 15, 20, 30, 40, 60, 80, 100, 120\},$$

we fitted a second-order trend-filtering model using the `genlasso` implementation in R.

For each fitted mean, we evaluated Gaussian, Poisson, and negative-binomial observation models. The combination of trend-filtering degrees of freedom and observation model was selected by the Akaike information criterion. The fitted mean from the selected combination was used as the downstream reconstruction target $\hat{\mu}$.

Let v_t denote the fitted conditional variance under the selected observation model. Thus,

$$v_t = \begin{cases} \hat{\sigma}^2, & \text{Gaussian,} \\ \hat{\mu}_t, & \text{Poisson,} \\ \hat{\mu}_t + \hat{\mu}_t^2 / \hat{\kappa}, & \text{negative binomial,} \end{cases}$$

where $\hat{\kappa}$ is the fitted negative-binomial dispersion parameter. For the frequency-constrained illustration, we summarized this possibly time-varying variance by the effective variance

$$\sigma_{\text{eff}}^2 = \frac{1}{T} \sum_{t=1}^T v_t.$$

This scalar variance provides a common quadratic scale for constructing and visualizing the locally non-identifiable perturbation set. It is not intended to replace the observation-model-specific bounds derived in Appendix S1.

S4.2 Delay kernel and discrete observation operator

The reconstruction used an empirical symptom-onset-to-case-report delay distribution for COVID-19 reported by

Zhang et al. The fitted continuous lognormal distribution was discretized into daily probabilities,

$$g_k = \Pr(k \leq \tau < k+1), \quad k = 0, \dots, 60,$$

and renormalized so that

$$\sum_{k=0}^{60} g_k = 1.$$

For an upstream incidence vector

$$U = (U_1, \dots, U_T)^\top,$$

the corresponding downstream mean is

$$KU,$$

where $K \in \mathbb{R}^{T \times T}$ is the lower-triangular causal convolution matrix with entries

$$K_{ts} = \begin{cases} g_{t-s}, & 0 \leq t-s \leq 60, \\ 0, & \text{otherwise.} \end{cases}$$

Equivalently,

$$(KU)_t = \sum_{k=0}^{\min(60, t-1)} g_k U_{t-k}.$$

The finite-series calculation therefore uses causal linear convolution rather than circular convolution.

S4.3 Frequency-constrained upstream reconstructions

To isolate the effect of assumptions about admissible temporal scales, we represented the upstream trajectory in a real Fourier basis. Let

$$\Phi_{f_c} = [\phi_1 \quad \dots \quad \phi_{q(f_c)}] \in \mathbb{R}^{T \times q(f_c)}$$

contain the normalized constant, sine, and cosine basis vectors whose frequencies satisfy

$$0 \leq f \leq f_c.$$

The basis vectors were normalized to have unit Euclidean norm. Apart from the constant component, each non-Nyquist Fourier frequency contributes one sine and one cosine basis vector.

An upstream trajectory in the cutoff-specific class is written as

$$U = \Phi_{f_c} \theta,$$

where $\theta \in \mathbb{R}^{q(f_c)}$ is the vector of Fourier coefficients. For each cutoff f_c , the reference reconstruction was obtained by solving

$$\hat{\theta}_{f_c} = \arg \min_{\theta} \left\{ \|K \Phi_{f_c} \theta - \hat{\mu}\|_2^2 + \lambda_{\theta} \sum_{j=2}^{q(f_c)} \theta_j^2 \right\},$$

subject to

$$\Phi_{f_c} \theta \geq 0.$$

The first coefficient, corresponding to the constant component, was excluded from the ridge penalty. We used the small numerical ridge parameter

$$\lambda_{\theta} = 10^{-6},$$

which stabilizes the optimization without materially determining the shape of the solution.

The resulting cutoff-specific reconstruction and its delayed projection are

$$U_{f_c}^* = \Phi_{f_c} \hat{\theta}_{f_c}, \quad D_{f_c}^* = K U_{f_c}^*.$$

We considered

$$f_c \in \left\{ \frac{1}{35}, \frac{1}{14}, \frac{1}{7} \right\} \text{ cycles/day.}$$

These cutoffs admit variation with periods no shorter than 35, 14, and 7 days, respectively. The three trajectories are therefore not estimates from three competing epidemiological models. They form a controlled sequence of reconstruction classes that progressively permits additional short-timescale structure.

S4.4 Frequency-constrained non-identifiable perturbation bands

For each cutoff-specific reference trajectory, we considered perturbations within the same Fourier subspace,

$$\delta U = \Phi_{f_c} a,$$

where $a \in \mathbb{R}^{q(f_c)}$ denotes the perturbation coefficients. Under the effective homoscedastic quadratic approximation used for this visualization, the downstream separability of a perturbation is

$$J(\delta U) = \frac{\|K \delta U\|_2^2}{\sigma_{\text{eff}}^2}.$$

To obtain an analytically interpretable frequency-domain approximation to this quantity, let

$$G_T(f) = \sum_{k=0}^{60} g_k \exp(-2\pi i f k)$$

denote the discrete Fourier response of the delay kernel evaluated on the length- T Fourier grid. For a normalized sine or cosine basis component at frequency f_j , define

$$\lambda_j = \frac{|G_T(f_j)|^2}{\sigma_{\text{eff}}^2}.$$

The separability of a perturbation is then approximated by the diagonal spectral quadratic form

$$J(\delta U) \approx a^\top \Lambda_{f_c} a, \quad \Lambda_{f_c} = \text{diag}(\lambda_1, \dots, \lambda_{q(f_c)}).$$

For a target total testing error $\alpha = 0.20$, the Gaussian non-identifiability threshold derived in Appendix S1 is

$$\tau_\alpha = 4(1 - \alpha)^2 = 2.56.$$

We therefore define the cutoff-specific perturbation ellipsoid

$$\mathcal{A}_{f_c} = \{a : a^\top \Lambda_{f_c} a \leq \tau_\alpha\}.$$

At each time t , the largest possible absolute perturbation within this ellipsoid is

$$B_{f_c}(t) = \sup_{a \in \mathcal{A}_{f_c}} |\phi_{f_c}(t)^\top a|,$$

where

$$\phi_{f_c}(t) = (\phi_1(t), \dots, \phi_{q(f_c)}(t))^\top.$$

By the Cauchy–Schwarz inequality in the Λ_{f_c} -weighted norm,

$$B_{f_c}(t) = \sqrt{\tau_\alpha \phi_{f_c}(t)^\top \Lambda_{f_c}^{-1} \phi_{f_c}(t)}.$$

Equivalently,

$$B_{f_c}(t) = \left[\tau_\alpha \sum_{j=1}^{q(f_c)} \frac{\phi_j(t)^2}{\lambda_j} \right]^{1/2}.$$

The band shown in Fig. 3A is

$$[\max\{U_{f_c}^*(t) - B_{f_c}(t), 0\}, U_{f_c}^*(t) + B_{f_c}(t)].$$

The lower boundary was truncated at zero to respect the nonnegativity of incidence.

The band is a pointwise envelope of the perturbation ellipsoid rather than a simultaneous confidence or credible band. In particular, the perturbation attaining the upper or lower endpoint may differ across time points. The band therefore visualizes the local geometry of the non-identifiable upstream set; it should not be interpreted as the sampling distribution of a specific reconstruction estimator.

S4.5 Epidemiological feature summaries

We evaluated three quantities derived from each cutoff-specific upstream trajectory: epidemic peak timing, epidemic peak magnitude, and the seven-day log growth rate. All feature summaries were calculated over the displayed interval from September 1, 2020, through March 31, 2021.

S4.5.1 Peak timing and peak magnitude

For each cutoff-specific reconstruction, we first formed a 14-day centered moving average,

$$\tilde{U}_{f_c}(t) = \frac{1}{14} \sum_{s \in \mathcal{W}_t} U_{f_c}^*(s),$$

where reflection padding was used at the boundaries of the finite interval. The peak index was defined as

$$t_{f_c}^* = \arg \max_t \tilde{U}_{f_c}(t).$$

For visualization, relative peak timing was calculated as

$$T_{f_c}^{\text{rel}} = \frac{t_{f_c}^*}{T_{\text{disp}} - 1},$$

where T_{disp} is the number of days in the displayed analysis interval.

The peak magnitude was evaluated from the unsmoothed cutoff-specific reconstruction at the peak index selected from the smoothed trajectory,

$$M_{f_c} = U_{f_c}^*(t_{f_c}^*).$$

Relative peak magnitude was then defined as

$$M_{f_c}^{\text{rel}} = \frac{M_{f_c}}{\max_{f'_c} M_{f'_c}}.$$

The common 14-day smoothing used to locate the epidemic peak reduces the sensitivity of the peak summaries to isolated daily fluctuations while preserving the broad epidemic-wave structure.

S4.5.2 Seven-day log growth rate

For each reconstructed upstream trajectory, we calculated the seven-day log growth rate

$$r_{7,f_c}(t) = \frac{1}{7} [\log \{U_{f_c}^*(t+7) + c\} - \log \{U_{f_c}^*(t) + c\}],$$

where $c = 1$ is a numerical offset used to avoid taking the logarithm of zero. The resulting growth-rate trajectory was smoothed using a five-day centered moving average with reflection padding at the boundaries. This final smoothing was applied identically to all cutoff-specific trajectories.

S4.6 Spectral sensitivity of the seven-day growth rate

The seven-day growth rate is nonlinear in $U(t)$, but its local spectral sensitivity can be seen through a first-order perturbation. Let

$$U_\epsilon(t) = U(t) + \epsilon \delta U(t).$$

Differentiating $r_7(t)$ with respect to ϵ at $\epsilon = 0$ gives

$$\left. \frac{\partial r_7[U_\epsilon](t)}{\partial \epsilon} \right|_{\epsilon=0} = \frac{1}{7} \left[\frac{\delta U(t+7)}{U(t+7)+c} - \frac{\delta U(t)}{U(t)+c} \right].$$

If the denominator varies slowly over a local interval, so that

$$U(t+7)+c \approx U(t)+c \approx q(t),$$

then

$$\delta r_7(t) \approx \frac{\delta U(t+7) - \delta U(t)}{7q(t)}.$$

The corresponding temporal-difference operator has frequency response

$$H_7(f) = \frac{e^{2\pi i 7f} - 1}{7},$$

with magnitude

$$|H_7(f)| = \frac{2|\sin(7\pi f)|}{7}.$$

For low frequencies,

$$|H_7(f)| \approx 2\pi|f|,$$

so the operator suppresses nearly constant components and increases its sensitivity as frequency moves away from zero. Relative to summaries of the overall level or broad epidemic-wave shape, the growth-rate calculation therefore places greater emphasis on short-timescale differences in the reconstructed trajectory.

The five-day moving average subsequently applied to the estimated growth rate attenuates the highest-frequency numerical variation, but it does not remove the underlying dependence of local growth estimates on upstream components that are more weakly transmitted through the delay process.

Peak timing and peak magnitude are nonlinear global functionals and do not possess a fixed linear transfer function analogous to $H_7(f)$. Their relative stability in Fig. 3 is therefore an empirical property of the admissible reconstruction family considered here, rather than a universal theorem that peak summaries are always identifiable. The analysis shows that, for this epidemic series and these reconstruction classes, the broad peak structure is considerably more stable than the local growth-rate trajectory.

S5 Controlled synthetic event-generation framework

This section describes the controlled synthetic framework used to construct the event-identifiability landscapes in Fig. 4. The objective of these simulations was not to reproduce a specific historical epidemic, but to isolate how different event parameterizations propagate through epidemic dynamics, delay filtering, and observational noise under a common epidemic setting.

S5.1 Baseline epidemic

All event classes were evaluated using the same baseline epidemic trajectory. The baseline incidence curve was defined as a smooth bimodal epidemic,

$$U_0(t) = 35 + 105 \exp\left[-\frac{(t-42)^2}{2 \times 16^2}\right] + 85 \exp\left[-\frac{(t-88)^2}{2 \times 23^2}\right], \quad (38)$$

evaluated over a simulation window consisting of 30 pre-event days and 120 post-event days. The event was centered at day 30.

The corresponding baseline reproduction-number trajectory was inferred from the renewal equation,

$$R_0(t) = \frac{U_0(t)}{\sum_k w_k U_0(t-k)}, \quad (39)$$

where w_k denotes the discrete generation-interval distribution. This baseline R_t trajectory served only as the reference transmission process on which event perturbations were imposed.

S5.2 Generation interval

Transmission dynamics followed the discrete renewal model described in the main text. The generation interval was represented by a Gamma distribution with mean 5 days and standard deviation 2 days, discretized over lags of 1–21 days and normalized to sum to one.

S5.3 Event parameterization

Each event modified the baseline reproduction number through

$$R(t; \theta) = R_0(t)m(t; \theta), \quad (40)$$

where

$$\theta = (a, \tau)$$

contains the peak proportional change in R_t (a) and the event duration (or rollout timescale) (τ).

Three stylized transmission perturbations were considered.

Mass gathering. A transient increase in transmission was represented as

$$m(t; \theta) = 1 + ah_\tau(t - t_0),$$

where h_τ is a smoothed finite-duration window normalized to have unit maximum.

Temporary lockdown. A transient reduction in transmission was represented as

$$m(t; \theta) = 1 - ah_\tau(t - t_0),$$

using the same normalized event window.

Vaccination rollout. Vaccination was represented by a monotone rollout curve,

$$m(t; \theta) = 1 - aC_\tau(t),$$

where $C_\tau(t)$ is a normalized logistic coverage curve increasing from 0 to 1 over the rollout period.

The event window h_τ was constructed using two opposing logistic transitions,

$$h_\tau(t) = \sigma\left(\frac{t - t_{\text{start}}}{e}\right) - \sigma\left(\frac{t - t_{\text{end}}}{e}\right),$$

where

$$\sigma(x) = \frac{1}{1 + e^{-x}},$$

followed by normalization to unit maximum.

The vaccination rollout curve was generated using a logistic function centered at the midpoint of the rollout period and linearly rescaled so that

$$C_\tau(t_{\text{start}}) = 0, \quad C_\tau(t_{\text{end}}) = 1.$$

S5.4 Renewal simulation

For every candidate event profile,

$$R(t; \theta) = R_0(t)m(t; \theta),$$

the corresponding incidence trajectory was generated recursively using

$$U(t; \theta) = R(t; \theta) \sum_k w_k U(t - k; \theta). \quad (41)$$

The first generation-interval window was initialized using the baseline trajectory. Consequently, within each event family every candidate profile was generated by perturbing the same baseline epidemic through the same renewal process. Differences across the event-identifiability landscape therefore arose solely from differences in the event parameterization under a common epidemic and observation setting.

S5.5 Observation model

The downstream conditional mean was obtained through convolution with a common reporting-delay distribution,

$$D(t; \theta) = p(U * g)(t),$$

where $p = 1$ in all controlled simulations.

The delay distribution was modeled by a discretized lognormal distribution with mean 7 days, standard deviation 6 days, and maximum delay 40 days.

Full linear convolution was retained throughout,

$$D = U * g,$$

rather than truncating the convolution to the original observation window. This preserves the exact Fourier relationship

$$\hat{D} = \hat{G}\hat{U}$$

after zero-padding and ensures exact agreement between the time-domain and spectral implementations of the separability functional.

S5.6 Negative-binomial observations

Illustrative observations were generated independently from

$$Y_t \sim \text{NB}(\mu_t, \kappa),$$

where

$$\mu_t = D(t; \theta),$$

and

$$\text{Var}(Y_t) = \mu_t + \frac{\mu_t^2}{\kappa}.$$

A common dispersion parameter

$$\kappa = 20$$

was used throughout all simulations.

One synthetic realization was generated from each reference event and displayed in Fig. 4 only to illustrate the scale of observational variability. All identifiability calculations were performed using the corresponding downstream conditional means rather than the sampled observations.

S5.7 Reference event profiles

The reference event profiles corresponded to a peak proportional 50% change in R_t .

The reference durations were

3 days, 10 days, 60 days,

for the mass gathering, temporary lockdown, and vaccination rollout, respectively.

Candidate event landscapes were evaluated over

$$a \in [0.05, 0.90]$$

and event-specific duration grids covering

1–30, 3–60, 10–120

days for the three event classes.

S5.8 Construction of the event-identifiability landscapes

For every point on the parameter grid,

$$\theta = (a, \tau),$$

the corresponding incidence trajectory was generated through the renewal model and mapped to the downstream conditional mean.

The event-identifiability landscape was then computed using

$$J_{\text{event}}(\theta, \theta^*) = \int \frac{|p\hat{G}(f)|^2 |\Delta\hat{U}(f)|^2}{N(f)} df,$$

where

$$\Delta U = U(\theta) - U(\theta^*).$$

Thus every point in the landscape represents the downstream separability between a candidate event profile and the specified reference profile.

S5.9 Selection of illustrative alternatives

For each event class, one representative alternative was selected automatically for visualization.

Candidates were restricted to those lying immediately outside the sufficient non-identifiability region,

$$J_{\text{threshold}} < J_{\text{event}} \leq 3J_{\text{threshold}},$$

while satisfying minimum parameter separation from the reference and excluding points near the parameter-space boundary.

Among the remaining candidates, the displayed example was chosen to maximize the relative upstream

root-mean-square difference from the reference trajectory. This procedure produces examples with substantial differences in transmission dynamics while remaining close to the theoretical non-identifiability boundary.

S6 Sensitivity Analysis

The preceding analyses treat the delay distribution $g(\tau)$ as known. In practice, epidemiological delay distributions are estimated from finite data and may be affected by sampling variability, right truncation, and reporting incompleteness. Uncertainty in the location or dispersion of the delay distribution changes its frequency response $|\hat{G}(f)|^2$, and therefore changes the amount of upstream spectral variation that remains distinguishable after delayed observation.

We examined how the single-frequency identifiability boundary varies across plausible delay distributions and under local misspecification of empirically estimated delay parameters. For an upstream perturbation with period T , the corresponding frequency is

$$f_T = \frac{1}{T}.$$

Under the additive Gaussian observation model used for this sensitivity analysis, the non-identifiability threshold is

$$\tau_\alpha = 4(1 - \alpha)^2.$$

The minimum required frequency-domain contrast-to-noise ratio for statistical distinguishability is therefore

$$Q_{\min}(T; g) = \frac{\tau_\alpha}{p^2 |\hat{G}(1/T)|^2}. \quad (42)$$

We set $\alpha = 0.20$, giving $\tau_\alpha = 2.56$, and fixed the observation probability at $p = 1$. Larger values of $Q_{\min}$ indicate that greater upstream spectral contrast relative to the effective downstream noise level is required for distinguishability.

We evaluated Eq. (42) for perturbation periods of 3, 7, and 14 days across a family of lognormal delay distributions. Each delay distribution was parameterized by its median delay and dispersion factor,

$$\text{median}(g) = \exp(\mu_{\log}), \quad d_g = \exp(\sigma_{\log}),$$

where $\mu_{\log}$ and $\sigma_{\log}$ are the mean and standard deviation of the logarithm of the delay. We parameterized each lognormal delay distribution by its median delay and dispersion factor, consistent with the parameterization used by Lessler et al. [?]. The resulting surfaces in Figures 5A–C show

$$\log_{10} Q_{\min}(T; g)$$

over the delay-parameter space.

The required contrast-to-noise ratio is generally larger for broader delay distributions and for shorter-period perturbations. This pattern follows directly from the stronger attenuation of high-frequency variation by dispersed delay kernels. The 3-day perturbation is therefore substantially more sensitive to the assumed delay distribution than the 7-day and 14-day perturbations. In some regions of parameter space, relatively modest changes in the delay parameters produce large multiplicative changes in the contrast required for distinguishability.

To place these theoretical surfaces in an epidemiological context, we overlaid representative empirical delay estimates. These included exposure-to-symptom-onset distributions and symptom-onset-to-report distributions from several infectious diseases. The empirical estimates occupy different regions of the delay-parameter space and consequently imply different levels of attenuation at the same perturbation frequency. Long and dispersed reporting delays tend to require substantially greater upstream contrast for distinguishing rapid variation than shorter and more concentrated delays.

We next quantified local sensitivity around each empirical delay estimate. Let m_0 and d_0 denote the original median delay and dispersion factor, respectively. For median misspecification, we evaluated

$$m = m_0(1 + \epsilon), \quad d = d_0,$$

whereas for dispersion misspecification we evaluated

$$m = m_0, \quad d = d_0(1 + \epsilon),$$

with

$$\epsilon \in [-0.30, 0.30].$$

Parameter combinations yielding an invalid lognormal dispersion factor, $d \leq 1$, were excluded.

For each perturbed delay distribution, we recomputed the theoretical single-frequency threshold in Eq. (42). Sensitivity was summarized relative to the threshold under the original empirical delay estimate:

$$\Delta \log_{10} Q_{\min} = \log_{10} \left[\frac{Q_{\min}(T; g_\epsilon)}{Q_{\min}(T; g_0)} \right], \quad (43)$$

where g_0 is the original delay distribution and g_ϵ is the distribution obtained after perturbing either its median or dispersion. Thus,

$$\Delta \log_{10} Q_{\min} = 0$$

corresponds to the original empirical estimate, positive values indicate a higher required contrast-to-noise ratio, and negative values indicate a lower required ratio.

Figures 5D–F show the effects of median-delay misspecification. The direction and magnitude of the response depend on both the empirical delay distribution

and the perturbation period. The largest changes generally occur for 3-day variation, whereas the required contrast for 14-day variation is comparatively less sensitive to moderate changes in the assumed median delay.

Figures 5G–I show the corresponding effects of dispersion misspecification. Increasing the dispersion factor generally increases the required contrast-to-noise ratio because a broader delay distribution more strongly attenuates rapid upstream variation. This effect is again most pronounced for the 3-day perturbation and progressively weaker for the 7-day and 14-day perturbations.

Overall, uncertainty in the delay distribution changes the quantitative location of the single-frequency identifiability boundary but does not alter the principal qualitative conclusion: rapid upstream variation is substantially more difficult to distinguish than slower variation after delayed observation. Inference concerning short-timescale epidemic dynamics is therefore especially sensitive to uncertainty in the assumed delay kernel and should be interpreted together with uncertainty in the estimated delay parameters.

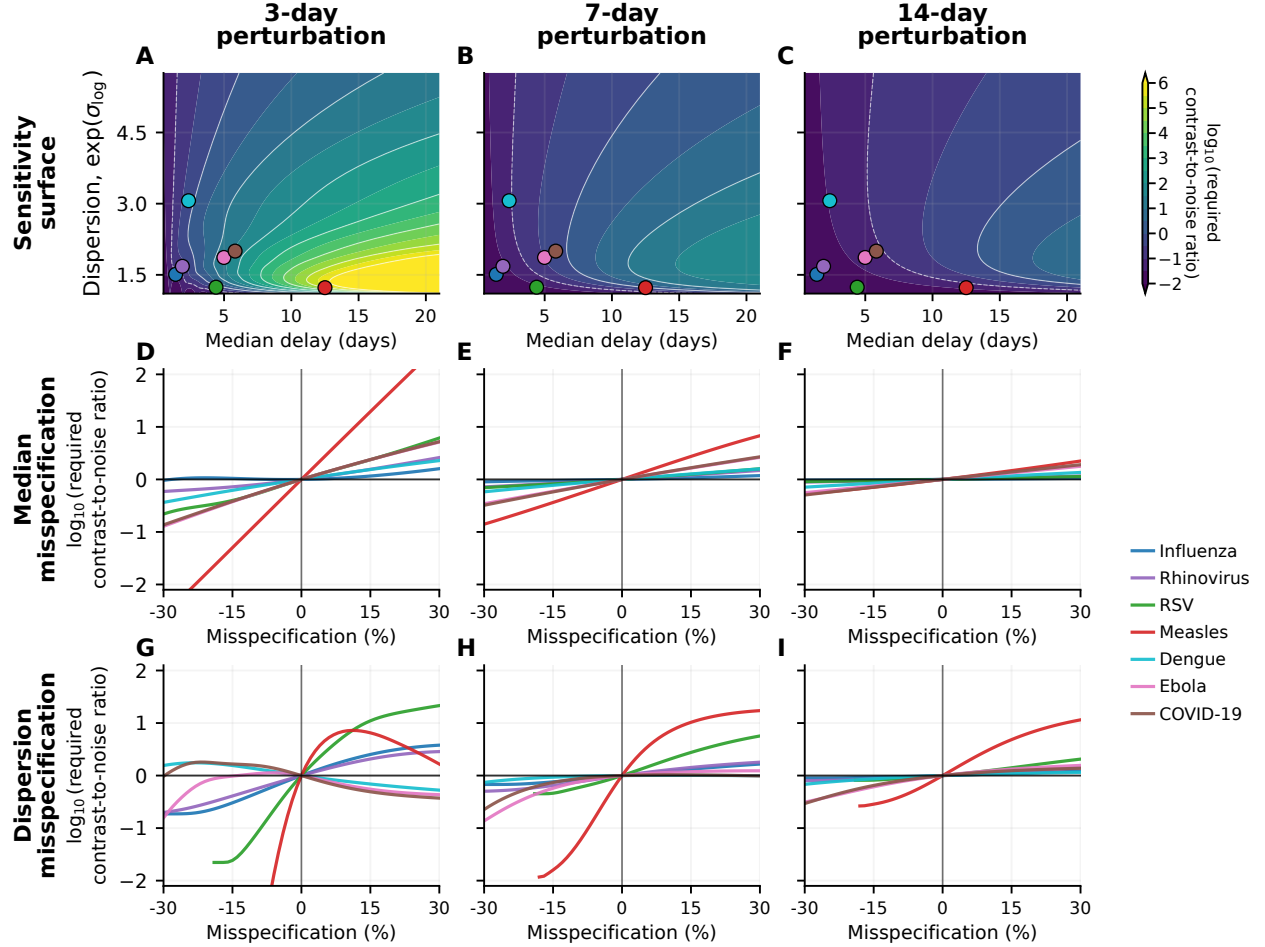


Figure 5: **Sensitivity of single-frequency identifiability to uncertainty in the delay distribution.** (A–C) Theoretical sensitivity surfaces showing $\log_{10} Q_{\min}$, where $Q_{\min} = 4(1 - \alpha)^2 / \{p^2 |\hat{G}(1/T)|^2\}$, across lognormal delay distributions parameterized by median delay and dispersion factor. Results are shown for perturbation periods T of 3, 7, and 14 days, with $\alpha = 0.20$ and $p = 1$. Colored points denote representative empirical exposure-to-onset and onset-to-report delay estimates. (D–F) Relative change in the required contrast-to-noise ratio after perturbing the median delay by up to $\pm 30\%$ while holding the dispersion factor fixed. Curves show $\log_{10}(Q_{\min}/Q_{\min,0})$, where $Q_{\min,0}$ is the value under the original empirical delay estimate. (G–I) Corresponding relative change after perturbing the dispersion factor by up to $\pm 30\%$ while holding the median delay fixed. Sensitivity is greatest for rapid perturbations and generally decreases as the perturbation period increases.